Bryant J. Curto

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Summary

PhD student building machine-checked correctness proofs for distributed systems in Rocq (formerly Coq), including proofs about Go implementations. Previously a back-end R&D engineer at Geopipe, Inc. (2018–2020), building its C++ pipeline for 3D city models.

Education

2022–present **PhD in Computer Science**, *Northeastern University*, Boston, MA
Expected graduation: Spring 2028; 4.0/4.0 GPA

2020–2022 **MMath in Computer Science**, *University of Waterloo*, Waterloo, ON
95.75/100.00 GPA

2014–2018 **BA CS with Honors & BA Math**, *New York University*, New York, NY
3.921/4.000 GPA; Summa Cum Laude; Phi Beta Kappa

Experience

2022–present **PhD Research Assistant**, *Northeastern University*, Boston, MA

- Lead developer of **Mosaic** (github.com/bryantcurto/consistency-semantics), a Rocq framework that verifies the **safety** (nothing bad happens) and **liveness** (something good eventually happens) of weakly consistent replicated storage system **Go implementations** (reads can return stale data) and of applications composing them. Mosaic extends Moveri (PLOS '25) beyond safety.
- Designed and implemented consistency models and safety reasoning around the key insight that consistency semantics form a partial order by strength, where stronger semantics compose from weaker ones (e.g., causal from the session guarantees). Compositional models of **16 consistency semantics** (e.g., eventual, PRAM) turn this into systematic proof reuse: each added semantics requires, on average, only **13.7% additional effort** (lines of spec and proof).
- Designed and directed the implementation of **liveness reasoning**—convergence (replicas progress towards storing the same state) and availability (each client eventually receives a response)—in linear temporal logic (LTL).
- Implemented and verified **three replicated storage systems in Go** with drastically different architectures (MOStore, Eiger, Causal Memory), and a **social media application** composing heterogeneous storage systems ([N] lines of Go; [N] lines of Rocq), with Claude writing the separation-logic (Iris) proofs. MOStore performs comparably to unverified Redis and achieves **21× the throughput** of verified Chapar.
- Uncovered implementation bugs violating safety and liveness (e.g., replicas misidentifying themselves and violating convergence), and hurting performance.

2020–2022 **Master's Research Assistant**, *University of Waterloo*, Waterloo, ON

Thesis: Improving Data Locality in Applications through Execution Delegation

- Built Accessprof, a profiling tool that predicts whether delegating a function's execution to another CPU core (via user-thread migration) improves data locality, by characterizing the application's working sets from memory accesses sampled with Linux perf. On the Apache HTTP server, it correctly predicted the effect for 3 of 4 system calls, and delegation improved throughput by up to **11%**.

2018–2020 **Back-End R&D Engineer**, *Geopipe, Inc.*, New York, NY

Summer 2017 **Back-End R&D Assistant**

Spring 2018

- Built the pipeline behind Geopipe's high-accuracy 3D city models (e.g., New York City, Boston), generated from LIDAR, satellite imagery, and OpenStreetMap data and licensed to game, simulation, and architecture customers; the company was later acquired by NBCUniversal.
- Wrote ~**200K lines** of its core in C++: ingesting raw input data; procedurally generating buildings, terrain, and decorations (e.g., windows); and fixing bugs.
- Provisioned and managed the AWS and Google Cloud servers that ran the pipeline to model new areas and update existing ones.
- Prototyped ML methods that predict model details (e.g., building shape, color, decoration) from raw input data (NSF SBIR-funded R&D).
- **First inventor** on a U.S. patent application for inferring objects from aerial imagery.
- Led 8 co-op interns, assigning and reviewing their work.

2016–2018 **Undergraduate Researcher**, *New York University*, New York, NY

- Showed how Spectre, a CPU flaw that leaks memory through cache side effects of speculative execution, can reveal a program's address-space layout, defeating address space layout randomization (ASLR) and re-enabling the attacks it defends against (e.g., return-oriented programming); senior honors thesis, *Ghosting ASLR: A Spectre Extension*.
- Implemented and evaluated a new database storage structure to reduce the latency of accesses that run concurrently with database entry migration; Courant Undergraduate Research Fellow.

Peer-Reviewed Publications

Curto, **Bryant J.**, Jeonghyeon Kim, Alan Wang, Gijung Im, Jieung Kim, Jeehoon Kang, and Ji-Yong Shin (2025). "Compositional Model-Driven Verification of Weakly Consistent Distributed Systems". In: *Proceedings of the 13th Workshop on Programming Languages and Operating Systems*. PLOS '25.

Xiong, Kezhi, Soonwon Moon, Joshua H Kang, **Bryant Curto**, Jieung Kim, and Ji-Yong Shin (2025). "ReCraft: Self-Contained Split, Merge, and Membership Change of Raft Protocol". In: *2025 55th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*.

Patents

Curto, **Bryant J**, Thomas Daniel Dickerson, and Daniel Christopher Ritchie (Mar. 2023). *Systems and methods for inferring object from aerial imagery*. US Patent App. 17/909,119.

Skills

Languages: Rocq (Coq), Go, Python, C++, C, Bash

Verification: Interactive theorem proving (Rocq); state machine refinement; safety; liveness; labeled transition system (LTS); linear temporal logic (LTL); compositional verification

Tools: Linux, Git, perf, GDB/LLDB, Goose, AI-assisted verification (Claude)

Additional

Talks: PLOS '25 (Seoul); New England Systems Verification Day ('24, '25); New England Systems Day ('25, '26)

Teaching: TA for Computer Systems (Northeastern, 2 semesters), Computer Networks (Waterloo, 2 terms), and Intro to CS 2 (Waterloo)